

Spinocerebellar ataxia 3 (SCA3) is a neurodegenerative disease that manifests as a progressive loss of coordination of the limbs, and frequently results in impaired speech and eye coordination. It is associated with extension of the polyglutamine (polyQ) region of the ataxin-3 protein. The polyQ region is encoded by a series of consecutive CAG codons known as poly-CAG repeats. These regions vary in length from individual to individual, ranging from as few as 10 to as many as 80 repeats. SCA3 has been reported in patients with polyQ regions exceeding ~55 repeats, and increased polyQ length correlates with earlier onset of the disease and greater clinical severity.

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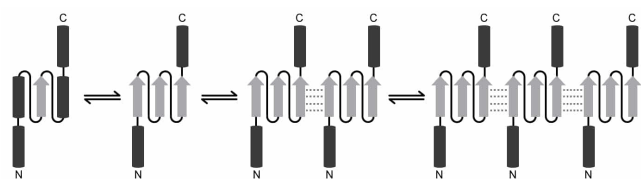


Figure 1 Depiction of the ataxin-3 polyQ region shifting from alpha-helical (cylinders) to beta-sheet (arrows) structure and aggregating. Dashed lines represent interactions between hydrophobic residues of adjacent ataxin-3 proteins.

Glutamine can be encoded by both CAG and CAA codons. Researchers investigated survival rates in fruit flies expressing ataxin-3 with polyQ regions of equal length encoded by either CAG-CAG or CAA-CAG repeats (Table 1). The protein products had identical amino acid sequences.

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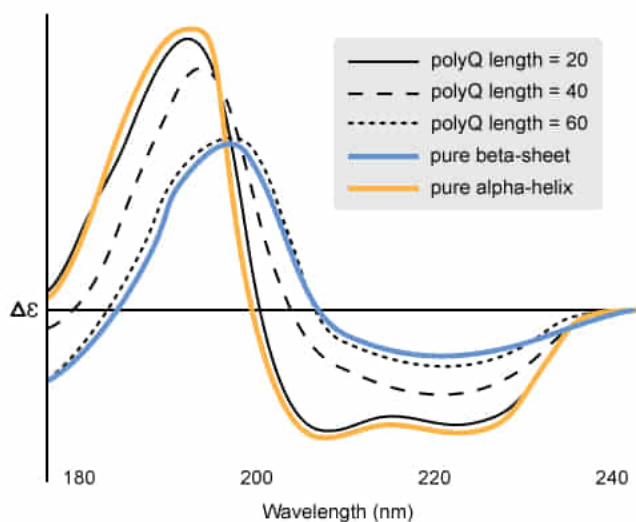
As the polyQ region increases in length, the resulting modification of ataxin-3 secondary structure would be observed as what change in the CD spectrum?

- ☐ A. An increase in the height of the peak at 190 nm [2%]
- ☐ B. Appearance of a peak at 190 nm and disappearance of the peak at 200 nm [3%]
- ✓ ☒ C. Disappearance of the peak at 190 nm and appearance of a peak at 200 nm [92%]
- ☐ D. A decrease in the height of the peak at 200 nm [1%]

Correct

92% answered correctly

Explanation:



Circular dichroism (CD) is a classic method of determining a protein's **secondary structure** based on its "ellipticity" $\Delta\epsilon$ at different wavelengths. According to the passage, alpha-helices have a maximum $\Delta\epsilon$ at 190 nm and beta-sheets have a maximum $\Delta\epsilon$ at 200 nm.

The passage also indicates that as the polyQ region of ataxin-3 increases in length, the protein **shifts from** a conformation **rich in alpha-helices to one rich in beta-sheets**. Therefore, the CD spectra should *lose* features that correspond to alpha-helices and *gain* features of beta-sheets as the length of the polyQ region increases. In a CD spectrum, this corresponds to the **disappearance** of

the peak at **190 nm** and the **appearance** of a peak at **200 nm**.

(Choice A) The height of the peak at 190 nm (alpha-helices) should *decrease* as the polyQ region length increases, indicating beta-sheet formation.

(Choice B) The peak at 190 nm (alpha-helices) should *disappear* as the polyQ region increases in length, and the peak at 200 nm (beta-sheets) should begin to appear.

(Choice D) The height of the peak at 200 nm should *increase* as beta-sheets form when the polyQ region increases in length.

Educational objective:

Circular dichroism reveals overall protein secondary structure, with the spectrum being a combination of structural features. The more alpha-helices a protein has, the more its spectrum will look like those of an alpha-helix; the more beta-sheets it has, the more its spectrum will look like those of a beta-sheet.

Time spent:0 sec

Subject:
Biochemistry

QID:400637

System:
Amino Acids and Proteins

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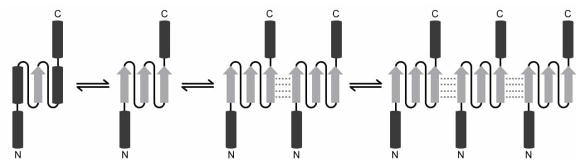


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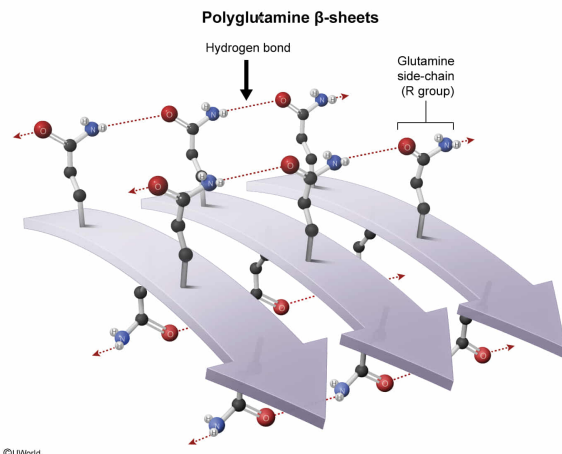
Within a single ataxin-3 protein, which of the following could increase the stability of the beta-conformation of long polyQ regions?

- ☐ A. Stacking interactions between glutamine side chains [18%]
- ☐ B. Frequent beta-turns induced by glutamine flexibility [7%]
- ☐ C. Rigidity of glutamine residues that induces tight turns [6%]
- ✔ ☒ D. Hydrogen bonding between glutamine side chains [67%]

Correct

68% answered correctly

Explanation:



In each strand of a beta-sheet, the amino acid side chains project out from and perpendicular to the backbone in alternating opposite directions. As beta-sheets form, the **R groups become aligned**. Glutamine is a polar uncharged amino acid with an amide side chain. Amides contain both a carbonyl oxygen and an amide -NH_2 group, which can act as a hydrogen bond acceptor and donor, respectively.

Because the side chains are aligned in beta-sheets and because the glutamine side chain can simultaneously **act as an acceptor and a donor**, polyQ beta-sheets can form **networks of hydrogen bonds**. In addition to the hydrogen bonds between the **peptide backbone** amide groups, the glutamine-mediated hydrogen bond networks *further* strengthen and stabilize the beta-sheet conformation, having a stronger effect as polyQ length increases.

(Choice A) Stacking interactions can indeed stabilize secondary structures. However, they occur primarily between **aromatic side chains** such as phenylalanine, tyrosine, and tryptophan.

(Choice B) It is **glycine**, not glutamine, that has the flexibility favorable for beta-turns.

(Choice C) The rigidity of **proline**, along with its tendency to adopt the *cis*-conformation, is conducive to tight turns. However, glutamine is not particularly rigid.

Educational objective:

Aligned glutamine side chains in adjacent strands of beta-sheets can form hydrogen bonds with each other because they can act as both donors and acceptors.

Polyglutamine beta-sheets can form networks of side-chain hydrogen bonds, adding considerable stability to the beta-structure.

Time spent:0 sec

Subject:
Biochemistry

QID:400638

System:
Amino Acids and Proteins

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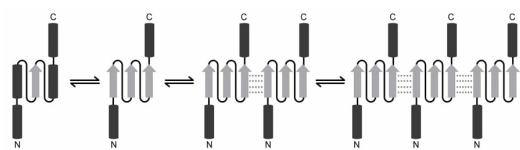


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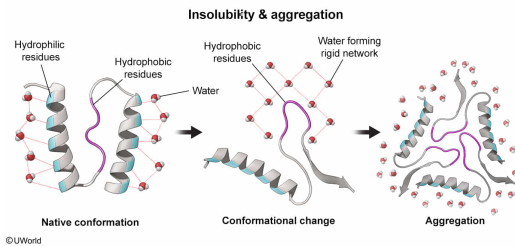
Why might ataxin-3 aggregates become insoluble?

- ☒ A. The change in conformation from alpha-helix to beta-sheet exposes hydrophobic residues. [55%]
- ☐ B. The conformational change disrupts side-chain ionic interactions between glutamine residues. [5%]
- ☐ C. Interactions between glutamine side chains exclude them from interacting with water. [21%]
- ☐ D. Beta-sheets expose more of the hydrophobic protein backbone than do alpha-helices. [17%]

Correct

55% answered correctly

Explanation:



Protein folding is primarily driven by the **hydrophobic effect**, which causes proteins to adopt a conformation that hides as many **hydrophobic residues in the interior** of the protein as possible. Changes that cause a protein to misfold often **expose hydrophobic residues** to the aqueous environment. Misfolded proteins often **group together** (ie, **aggregate**) to hide these exposed residues. Aggregation often results in a significant **decrease in the solubility** of the protein.

The passage states that ataxin-3 proteins aggregate when they contain large polyQ regions, and that these aggregated proteins have decreased alpha-helical character and increased beta-sheet character. This result suggests that as the alpha-helices of ataxin-3 are converted to beta-sheets, the protein undergoes an overall **conformational change** that exposes hydrophobic residues to water. This in turn likely causes the hydrophobic residues of multiple proteins to aggregate to the exclusion of water, which decreases the solubility of ataxin-3.

(Choice B) **Glutamine** has an uncharged polar side chain. It has no ionic interactions to disrupt.

(Choice C) The interaction of glutamine side chains with each other does not prevent their interaction with water. Hydrogen bonds between side chains can transiently break to bond with water, and vice versa.

(Choice D) **Peptide backbones** are not very hydrophobic because they have both a carbonyl and an amino group that can form hydrogen bonds.

Educational objective:

Proteins that adopt non-native conformations are generally forced to expose more hydrophobic residues to the aqueous environment. These residues cannot interact with water, and therefore tend to aggregate to minimize exposure due to the hydrophobic effect. The inability to interact with water results in precipitation out of solution.

Time spent: 0 sec

Subject:
Biochemistry

QID: 400639

System:
Amino Acids and Proteins

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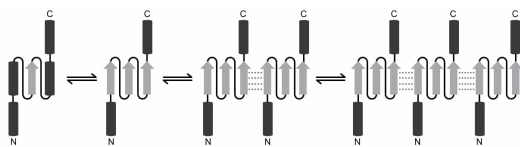


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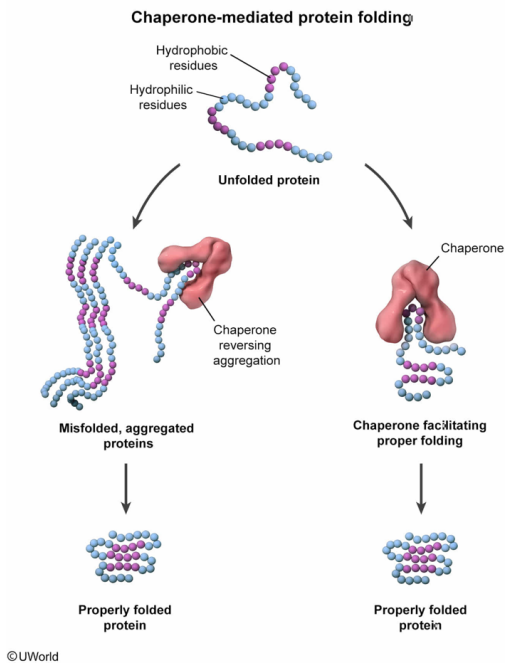
Researchers hypothesized that overexpression of the chaperone protein HSP70 could reduce toxicity associated with ataxin-3 containing long polyQ regions. Which line of reasoning might have led them to this hypothesis?

- ☐ A. HSP70 may downregulate transcription of additional ataxin-3. [5%]
- ☒ B. HSP70 may facilitate proper folding and solubility of ataxin-3. [74%]
- ☐ C. HSP70 may proteolytically cleave amyloid fibers, removing them from the system. [9%]
- ☐ D. HSP70 may disrupt hydrogen bonds between beta-sheets of adjacent proteins. [9%]

Correct

75% answered correctly

Explanation:



Molecular chaperones are defined as proteins that **facilitate the proper folding** of other proteins. They usually accomplish this by binding hydrophobic regions of nascent, misfolded, or aggregated proteins, preventing exposure of those regions to aqueous solvent. This arrangement prevents incorrect interactions between hydrophobic residues and can **prevent or reverse aggregation** by blocking interactions between hydrophobic regions of separate polypeptides.

The passage indicates that misfolding and aggregation of ataxin-3 is associated with toxicity. Because HSP70 is a **chaperone**, it is reasonable to hypothesize that it could diminish the toxic effects of misfolded and aggregated proteins such as ataxin-3 by helping them fold correctly.

(Choice A) HSP70 would have to act as a negative transcription factor, not a chaperone, to downregulate ataxin-3 transcription.

(Choice C) Although chaperones can facilitate entry of misfolded proteins into degradation pathways, they have no proteolytic activity of their own.

(Choice D) Disruption of the interactions between beta-sheets could help disaggregate amyloid fibers, which may increase solubility and reduce toxicity. However, the Figure 1 caption indicates that the interactions *between* ataxin-3 proteins are facilitated by hydrophobic residues, so HSP70 would not disrupt hydrogen bonds between adjacent sheets.

Educational objective:

Chaperones are a class of proteins that help other proteins fold correctly. They decrease the tendency of a protein to aggregate and increase its solubility.

Time spent: 0 sec

Subject:
Biochemistry

QID: 400640

System:
Amino Acids and Proteins

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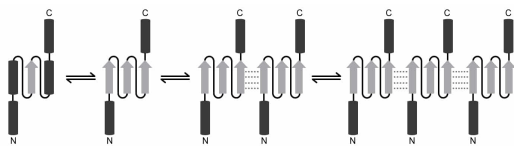


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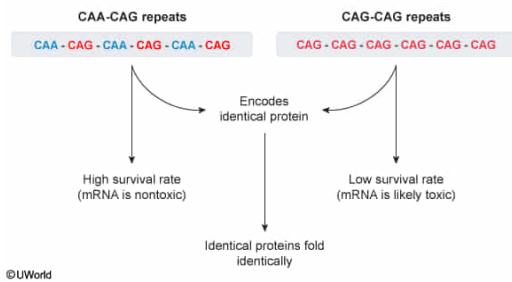
Scientists concluded that ataxin-3 folding is not the only factor that contributes to SCA3. Is this conclusion supported by the data?

- ✔ ☒ A. Yes; flies with identical ataxin-3 proteins but distinct ataxin-3 mRNA had different survival rates. [85%]
- ☐ B. Yes; flies with different ataxin-3 expression and aggregation levels had the same survival rates. [3%]
- ☐ C. No; flies with more ataxin-3 expression had a lower survival rate. [6%]
- ☐ D. No; flies with less ataxin-3 aggregation had a higher survival rate. [4%]

Correct

86% answered correctly

Explanation:



The **genetic code** is **degenerate**, meaning **multiple codons** can encode the **same amino acid**. Both CAA and CAG encode glutamine (Q), so genes containing CAG-CAG or CAA-CAG repeats produce identical proteins with polyQ regions of the same length. Identical proteins are likely to fold identically. This characteristic is supported by the data in Table 1, which show that aggregation (a result of abnormal folding) occurs at similar levels for ataxin-3 encoded by CAG-CAG or CAA-CAG repeats. Nevertheless, flies with CAG-CAG repeats had a much lower survival rate than flies with CAA-CAG repeats, indicating that at least one other factor, such as the mRNA composition, contributes to SCA3 symptoms.

(Choice B) The flies assessed in Table 1 had similar ataxin-3 expression and aggregation levels but vastly different survival rates.

(Choice C) The flies with the lower survival rate (CAG-CAG) expressed ataxin-3 at slightly *lower* levels than the flies with a higher survival rate, so increased ataxin-3 expression did not cause lower survival rates.

(Choice D) CAG-CAG and CAA-CAG flies had similar levels of ataxin-3 aggregation, but the flies with the higher survival rate (CAA-CAG) had a slightly *higher* aggregation level.

Educational objective:

Identical proteins may be encoded by genes with different codon sets because the genetic code is degenerate. Proteins expressed by different mRNAs with codons that code for the same amino acid sequence fold identically and are indistinguishable from each other.

Time spent: 0 sec

Subject:
Biochemistry

QID: 400641

System:
Amino Acids and Proteins